

The Abstractions Are Slipping

Four OS Mismatches in the Agent Era

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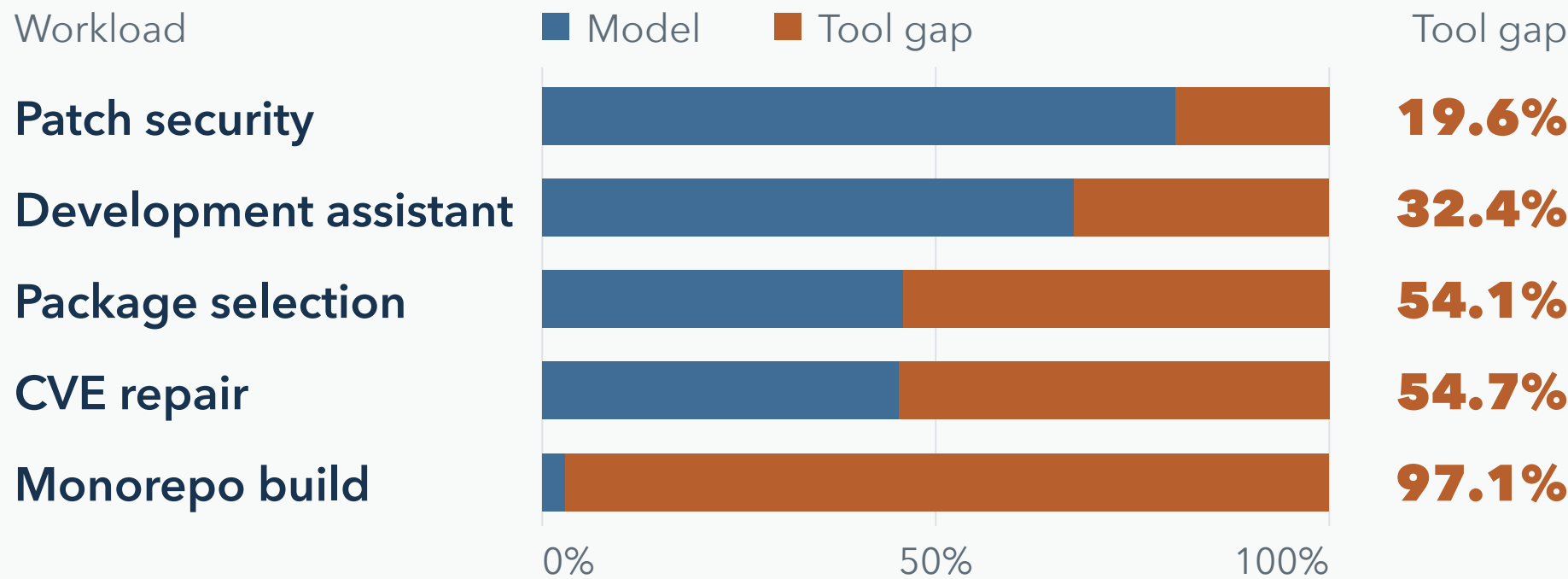
OS abstractions should evolve with agent execution.

RESEARCH GOAL Identify OS-visible agent semantics and the abstractions they need.

Agent work spans model calls, tools and waiting.

ACTIVE TIME • FIVE WORKLOADS

697 sessions · 4,715 model calls [2]



ELAPSED SESSION TIME

One production instance [1]

87% open / idle



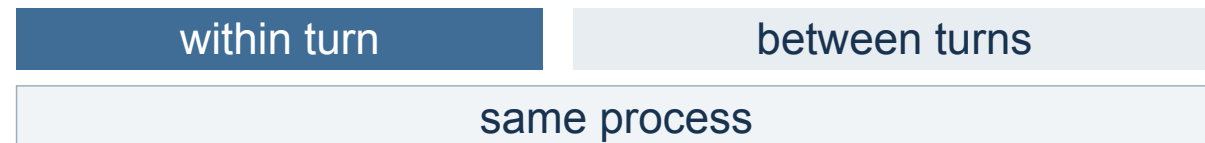
347 sessions · 2,682 model calls
3 weeks · rounded shares

Session lifetime ≠ active execution.

[1] Paper, §2 · [2] AgentSight summaries · expanded sample beyond the paper

Where do the abstractions stop fitting?

M1 Turns vs. processes



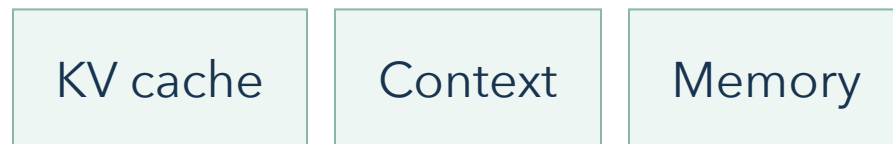
Process state does not name a turn.

M2 Tool contracts vs. wrappers



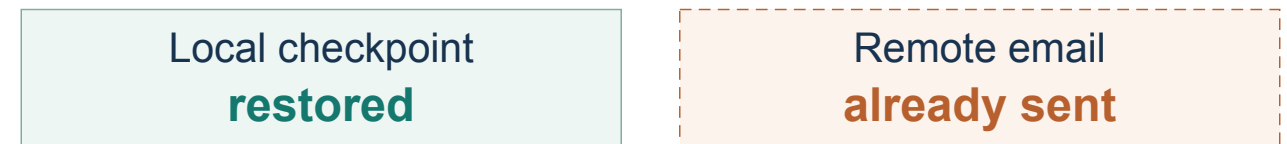
The contract need not follow the tool.

M3 State lifetimes vs. files



Different lifetimes. Different owners.

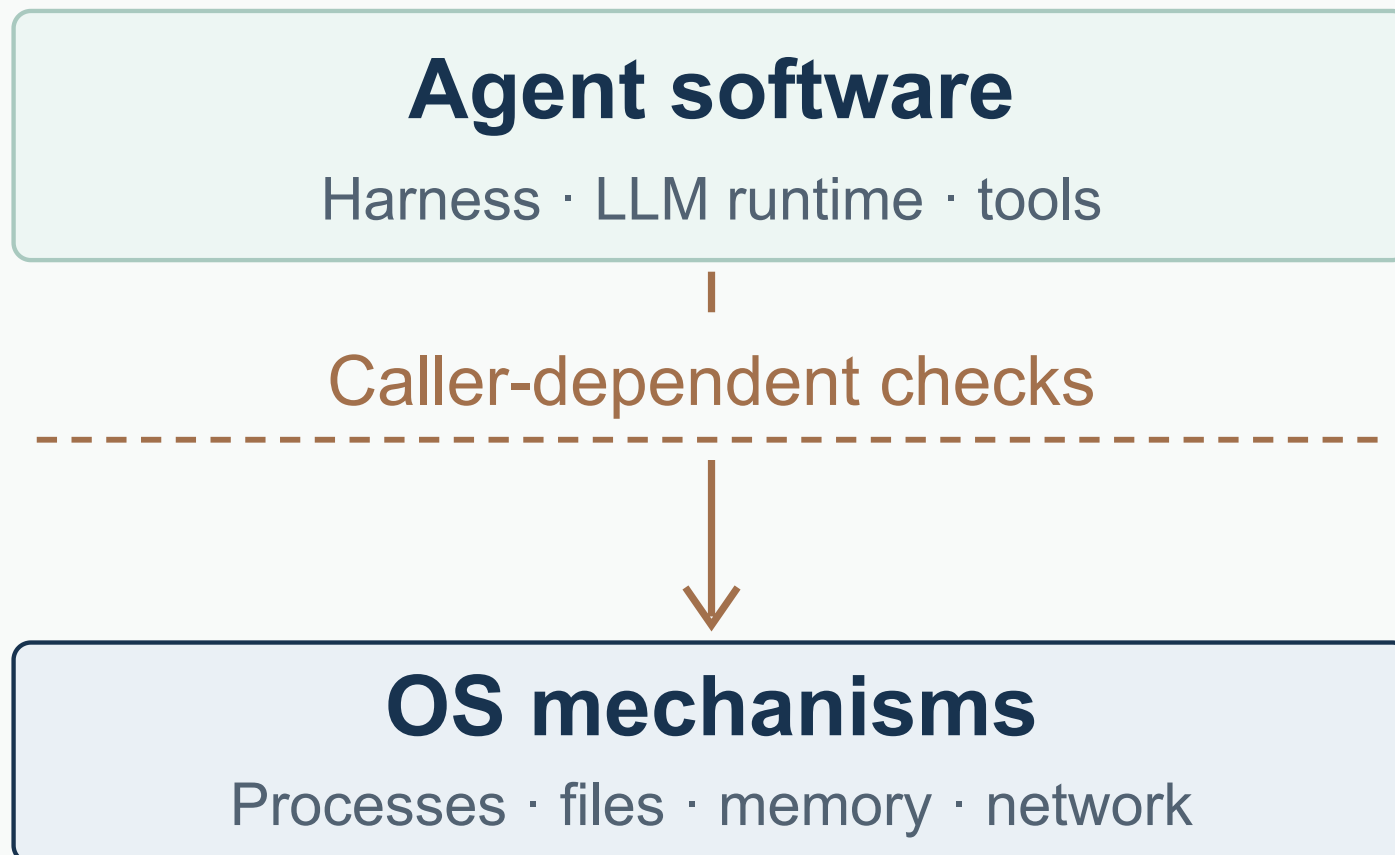
M4 Branches vs. external effects



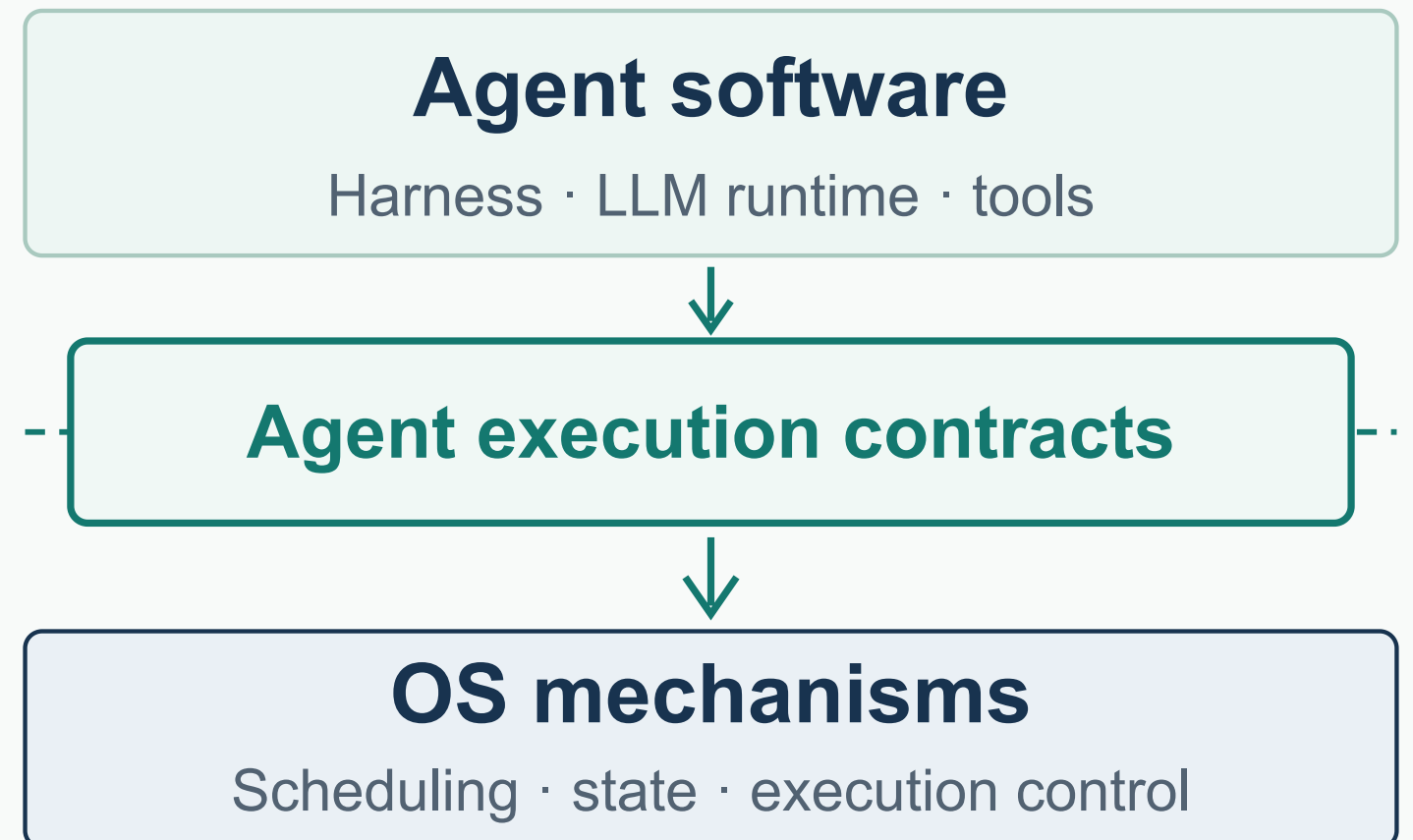
Local rollback cannot unsend an email.

Connect agent semantics to OS mechanisms.

Framework-local contracts



Shared system contracts



Research goal: identify OS-visible agent semantics and the abstractions they need.

Two contracts to begin.

Harnesses → OS

Signal turn boundaries.

Tools → OS

Declare side effects.

OS abstractions should evolve with agent execution.

Which agent semantics belong below the agent?



ANOLISA · code